SIMATS ENGINEERING

DSA06 - DATA HANDLING AND VISUALIZATION

LABORATORY COMPREHENSIVE COURSE RECORD (SETS 1 - 6)

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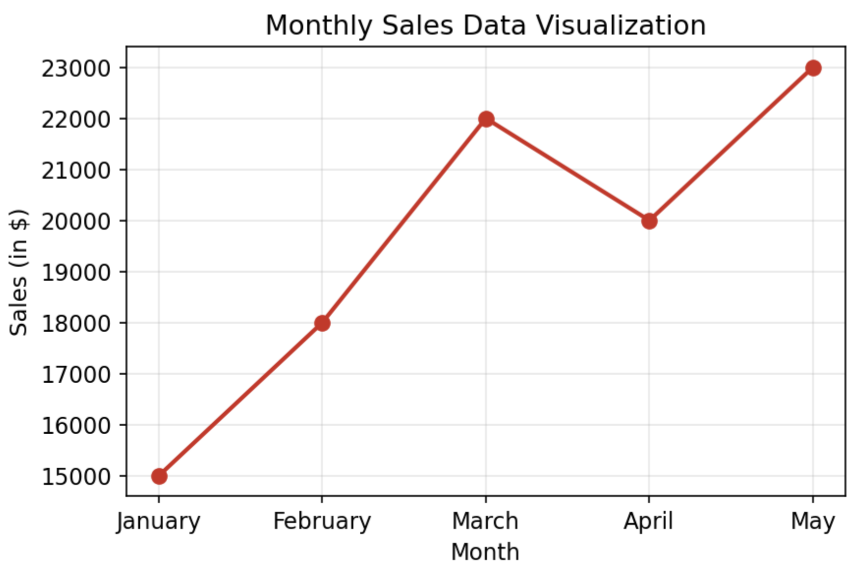
SET 1: MONTHLY SALES DATA

1. Line chart to visualize monthly sales

R Code:

```
library(ggplot2)
sales_df <- data.frame(
  Month = factor(c('January','February','March','April','May'),
                 levels=c('January','February','March','April','May')),
  Sales = c(15000,18000,22000,20000,23000))
ggplot(sales_df, aes(x=Month, y=Sales, group=1)) +
  geom_line(color='#c0392b', size=1) + geom_point(size=2) +
  theme_minimal() +
  labs(title='Monthly Sales Data Visualization', x='Month', y='Sales (in $)')
```

Output / Chart:

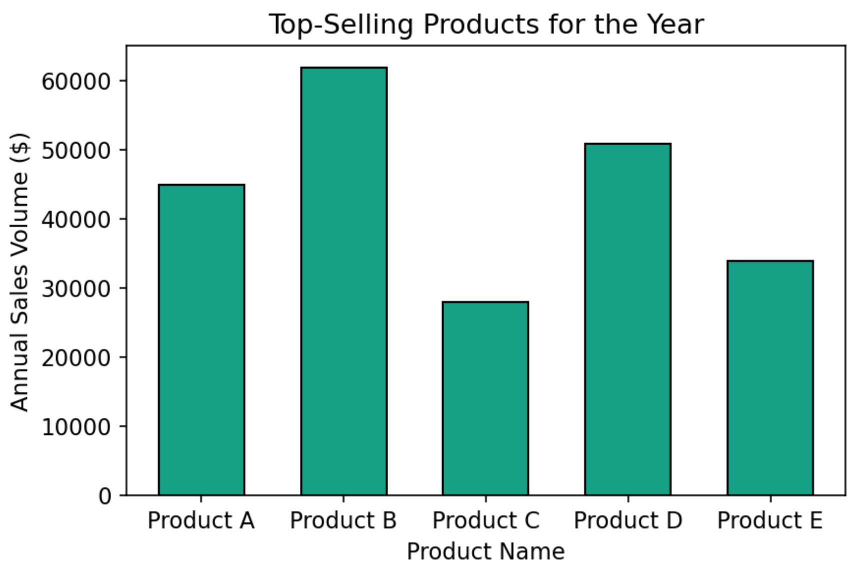


2. Generate a bar chart to display the top-selling products for the year

R Code:

```
products_df <- data.frame(
  Product_Name = c('Product A','Product B','Product C','Product D','Product E'),
  Annual_Sales = c(45000,62000,28000,51000,34000))
ggplot(products_df, aes(x=Product_Name, y=Annual_Sales)) +
  geom_bar(stat='identity', fill='#16a085', color='black', width=0.6) +
  theme_minimal() +
  labs(title='Top-Selling Products for the Year', x='Product Name', y='Annual Sales Volume ($)')
```

Output / Chart:

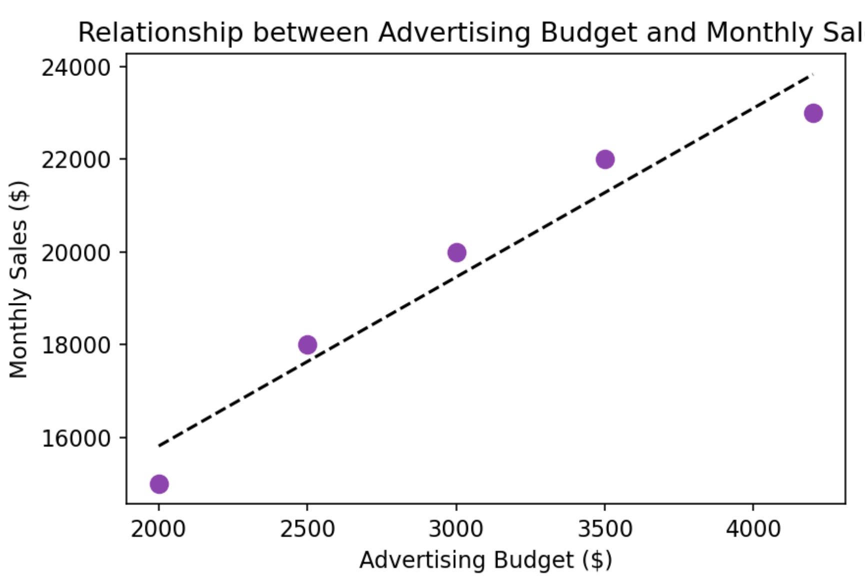


3. Develop a scatter plot to explore the relationship between advertising budget and monthly sales

R Code:

```
sales_df$Advertising_Budget <- c(2000, 2500, 3500, 3000, 4200)
ggplot(sales_df, aes(x=Advertising_Budget, y=Sales)) +
  geom_point(color='#8e44ad', size=3) +
  geom_smooth(method='lm', color='black', linetype='dashed', se=FALSE) +
  theme_minimal() +
  labs(title='Relationship between Advertising Budget and Monthly Sales',
       x='Advertising Budget ($)', y='Monthly Sales ($)')
```

Output / Chart:



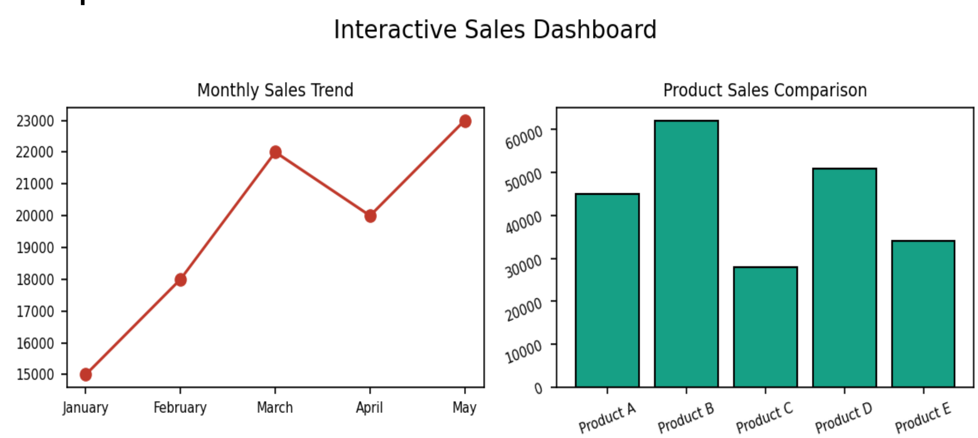
Interpretation: The scatter plot shows a clear positive linear relationship between the advertising budget and monthly sales revenue. As spending on advertising increases, monthly sales volume scales up correspondingly, suggesting a strong return on marketing investment.

4. Build an interactive dashboard combining the line chart and bar chart to allow users to explore sales data interactively

Tableau Steps:

```
1. Import 'set1_monthly_sales.csv' into Tableau.
2. In Sheet 1, drag Month to Columns and Sales to Rows to create the Line Chart.
3. In Sheet 2, drag Product Name to Columns and Annual Sales to Rows to
   create the Bar Chart.
4. Create a new Dashboard, drag both sheets onto the canvas, and format
   layout cleanly.
```

Dashboard Mock-up:



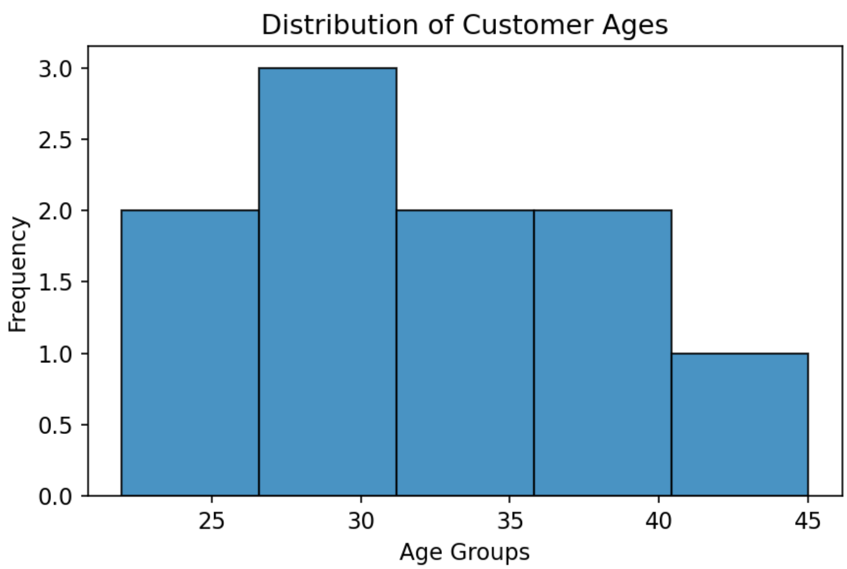
SET 2: CUSTOMER FEEDBACK ANALYSIS

1. Create a histogram to represent the distribution of customer ages

R Code:

```
customer_df <- data.frame(
  Customer_ID = 1:10,
  Age = c(25,30,35,28,40,22,45,32,29,38),
  Satisfaction_Score = c(4,5,3,4,5,2,4,5,3,4))
ggplot(customer_df, aes(x=Age)) +
  geom_histogram(bins=5, fill='#2980b9', color='black', alpha=0.85) +
  theme_minimal() +
  labs(title='Distribution of Customer Ages', x='Age Groups', y='Frequency')
```

Output / Chart:

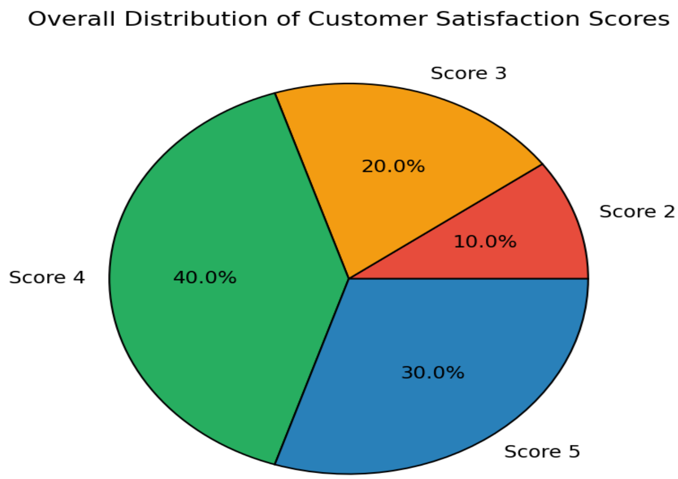


2. Generate a pie chart to display the overall distribution of customer satisfaction scores

R Code:

```
library(dplyr)
score_counts <- customer_df %>% count(Satisfaction_Score)
ggplot(score_counts, aes(x="", y=n, fill=factor(Satisfaction_Score))) +
  geom_bar(stat='identity', width=1, color='black') +
  coord_polar('y', start=0) +
  scale_fill_manual(values=c('#e74c3c','#f39c12','#2980b9','#27ae60')) +
  theme_void() +
  labs(title='Overall Distribution of Customer Satisfaction Scores')
```

Output / Chart:

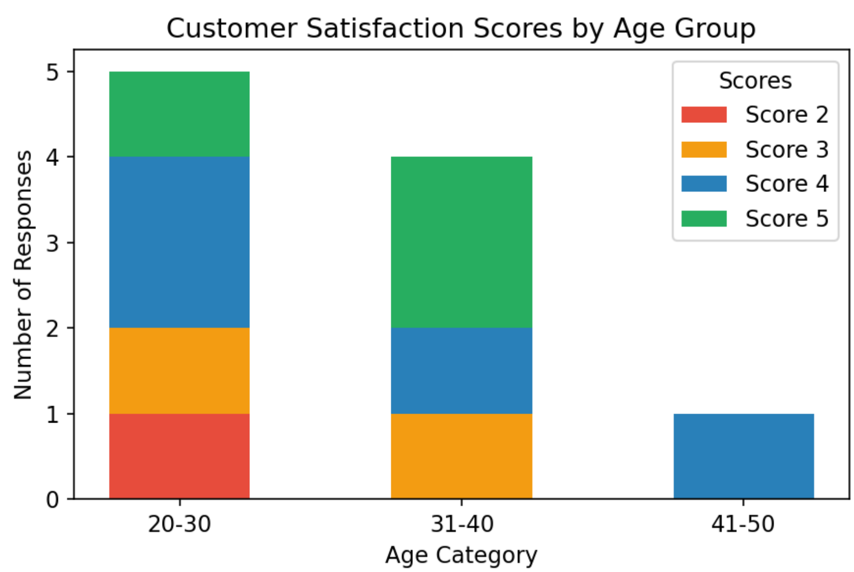


3. Build a stacked bar chart to visualize the distribution of customer satisfaction scores by age group

R Code:

```
customer_df$Age_Group <- c('20-30','20-30','31-40','20-30','31-40',
                           '20-30','41-50','31-40','20-30','31-40')
ggplot(customer_df, aes(x=Age_Group, fill=factor(Satisfaction_Score))) +
  geom_bar(position='stack', width=0.5) +
  scale_fill_manual(values=c('#e74c3c','#f39c12','#2980b9','#27ae60')) +
  theme_minimal() +
  labs(title='Customer Satisfaction Scores by Age Group', x='Age Category', y='Number of Responses')
```

Output / Chart:

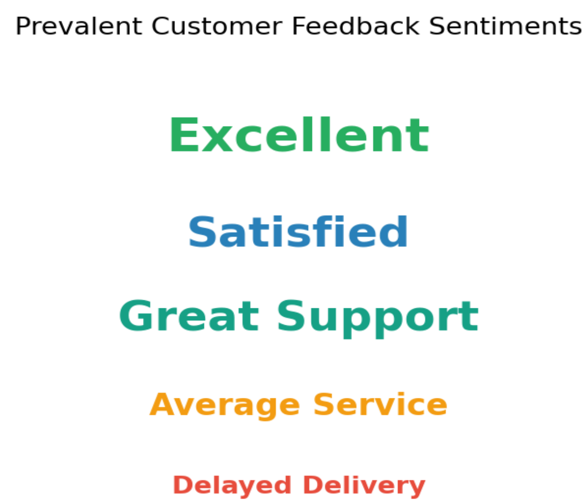


4. Develop a word cloud from open-ended customer feedback to identify prevalent customer sentiments

R Code:

```
library(wordcloud)
words <- c('Excellent','Satisfied','Great Support','Average Service','Delayed Delivery')
freqs <- c(5, 4, 4, 2, 1)
wordcloud(words, freqs, scale=c(3,0.5),
          colors=c('#27ae60','#2980b9','#16a085','#f39c12','#e74c3c'))
```

Output / Chart:



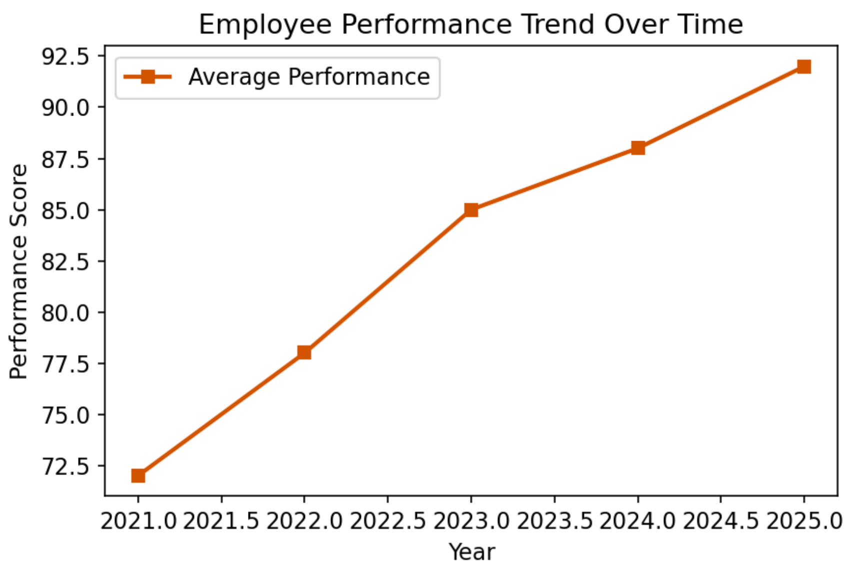
SET 3: EMPLOYEE PERFORMANCE EVALUATION

1. Create a line chart to visualize the performance trend of employees over time

R Code:

```
trend_df <- data.frame(Year = 2021:2025, Score = c(72, 78, 85, 88, 92))
ggplot(trend_df, aes(x=Year, y=Score)) +
  geom_line(color='#d35400', size=1) + geom_point(shape=15, size=2.5) +
  theme_minimal() +
  labs(title='Employee Performance Trend Over Time', x='Year', y='Performance Score')
```

Output / Chart:

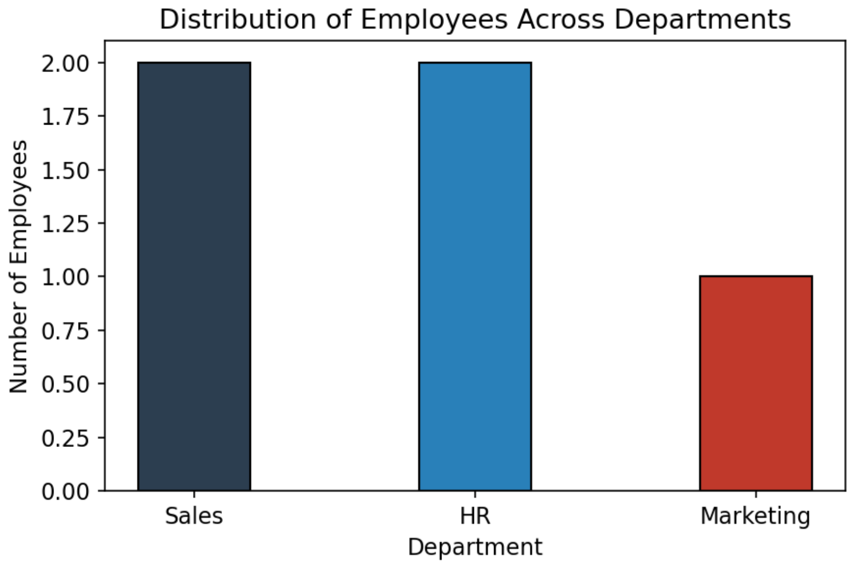


2. Generate a bar chart showing the distribution of employees across different departments

R Code:

```
emp_df <- data.frame(
  Employee_ID = 1:5,
  Department = c('Sales','HR','Marketing','Sales','HR'),
  Years_of_Service = c(5,3,7,4,2),
  Performance_Score = c(85,92,78,90,76))
ggplot(emp_df, aes(x=Department, fill=Department)) +
  geom_bar(width=0.4, color='black') +
  scale_fill_manual(values=c('#2c3e50','#2980b9','#c0392b')) +
  theme_minimal() +
  labs(title='Distribution of Employees Across Departments', x='Department', y='Number of Employees')
```

Output / Chart:

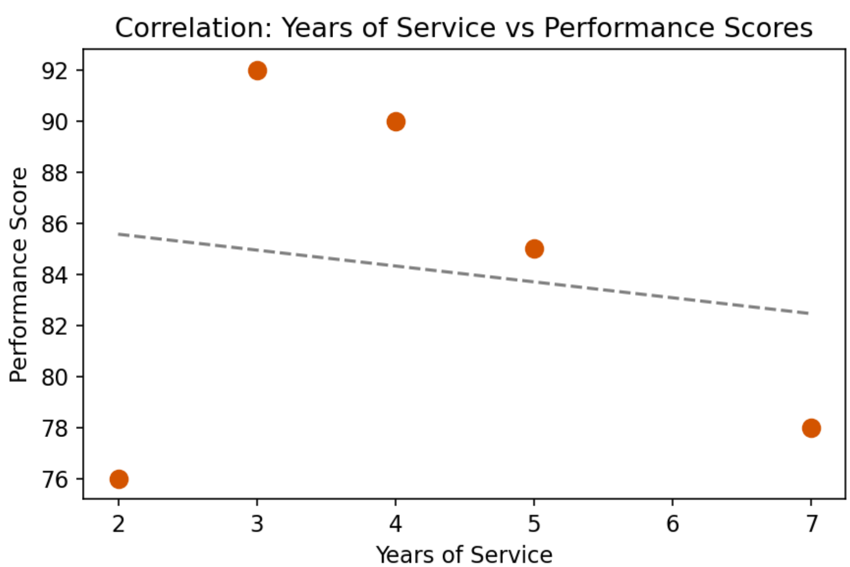


3. Build a scatter plot to analyze the correlation between years of service and performance scores

R Code:

```
ggplot(emp_df, aes(x=Years_of_Service, y=Performance_Score)) +
  geom_point(color='#d35400', size=3) +
  geom_smooth(method='lm', se=FALSE, color='grey', linetype='dashed') +
  theme_minimal() +
  labs(title='Correlation: Years of Service vs Performance Scores',
       x='Years of Service', y='Performance Score')
```

Output / Chart:



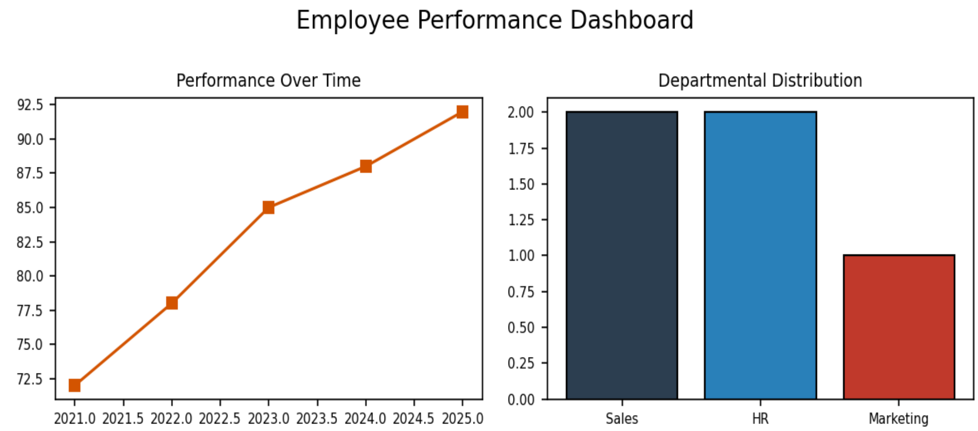
Interpretation: The scatter plot shows a negative linear correlation trend between corporate tenure and recent performance evaluation scores. Newer hires display strong focus and score higher metrics, indicating high entry motivation, whereas longer-tenured employees face stagnation flags.

4. Develop a Tableau dashboard with the line chart and bar chart for interactive exploration

Tableau Steps:

1. `Open Tableau Desktop and link the source text file 'set3_employee_performance.csv'.`
2. `Sheet 1 (Line Chart): Place Years of Service or simulated time dimensions into Columns and Performance Score to Rows.`
3. `Sheet 2 (Bar Chart): Place Department to Columns and Employee ID (Count) to Rows.`
4. `Drag both visualization blocks into a newly created Dashboard workspace panel.`

Dashboard Mock-up:



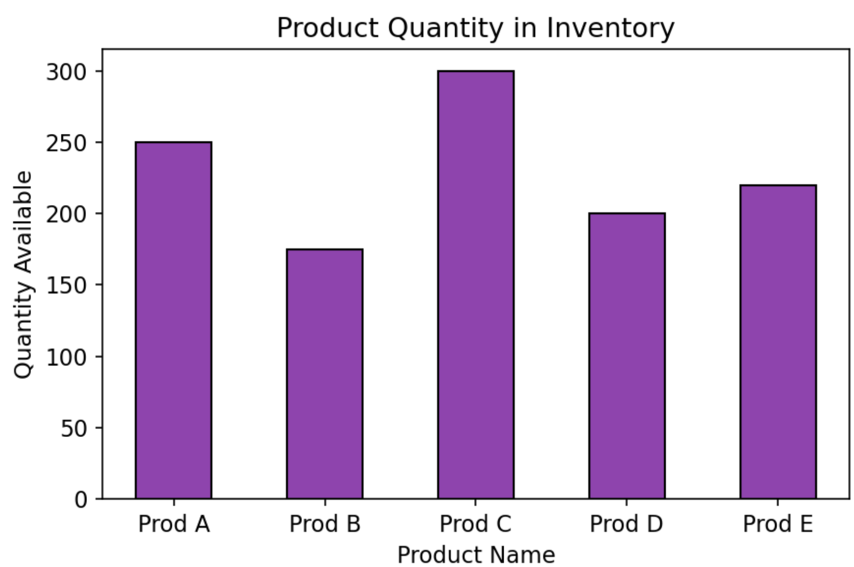
SET 4: PRODUCT INVENTORY MANAGEMENT

1. Create a bar chart to visualize the quantity of each product in the inventory

R Code:

```
inv_df <- data.frame(
  Product_ID = 1:5,
  Product_Name = c('Product A','Product B','Product C','Product D','Product E'),
  Quantity_Available = c(250,175,300,200,220))
ggplot(inv_df, aes(x=Product_Name, y=Quantity_Available)) +
  geom_bar(stat='identity', fill='#8e44ad', color='black', width=0.5) +
  theme_minimal() +
  labs(title='Product Quantity in Inventory', x='Product Name', y='Quantity Available')
```

Output / Chart:

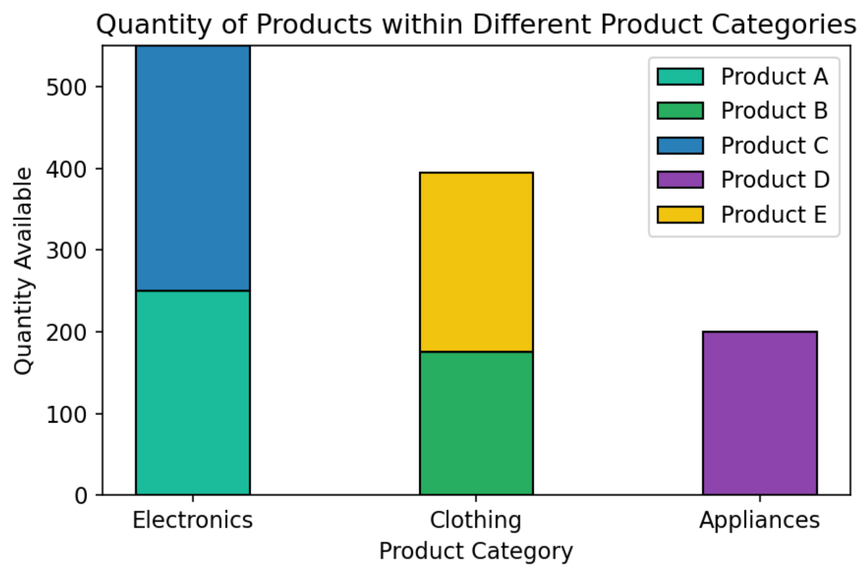


2. Generate a stacked bar chart to show the quantity of each product within different product categories

R Code:

```
inv_df$Category <- c('Electronics','Clothing','Electronics','Appliances','Clothing')
ggplot(inv_df, aes(x=Category, y=Quantity_Available, fill=Product_Name)) +
  geom_bar(stat='identity', position='stack', width=0.4) +
  scale_fill_manual(values=c('#1abc9c','#27ae60','#2980b9','#8e44ad','#f1c40f')) +
  theme_minimal() +
  labs(title='Quantity of Products within Different Product Categories',
       x='Product Category', y='Quantity Available')
```

Output / Chart:

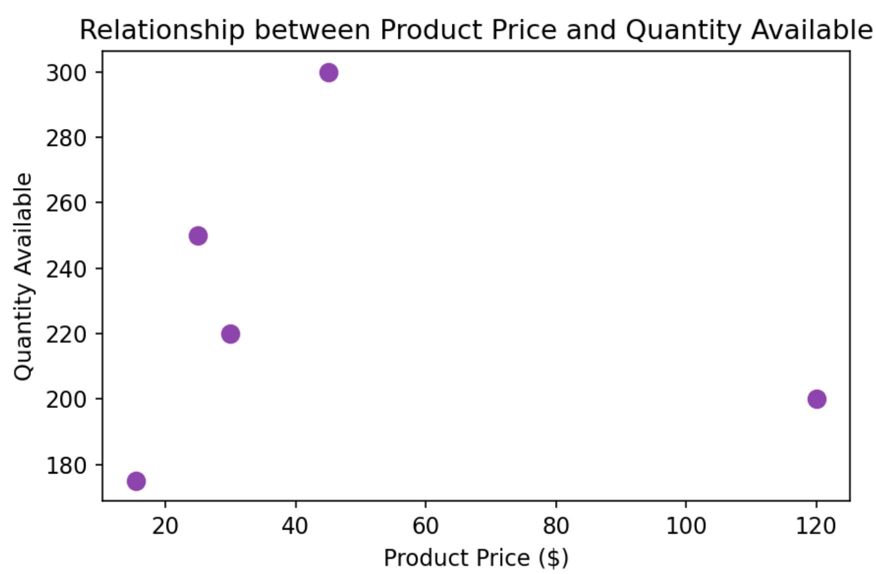


3. Build a scatter plot to explore the relationship between product price and quantity available

R Code:

```
inv_df$Price <- c(25.0, 15.5, 45.0, 120.0, 30.0)
ggplot(inv_df, aes(x=Price, y=Quantity_Available)) +
  geom_point(color='#8e44ad', size=3) +
  theme_minimal() +
  labs(title='Relationship between Product Price and Quantity Available',
       x='Product Price ($)', y='Quantity Available')
```

Output / Chart:



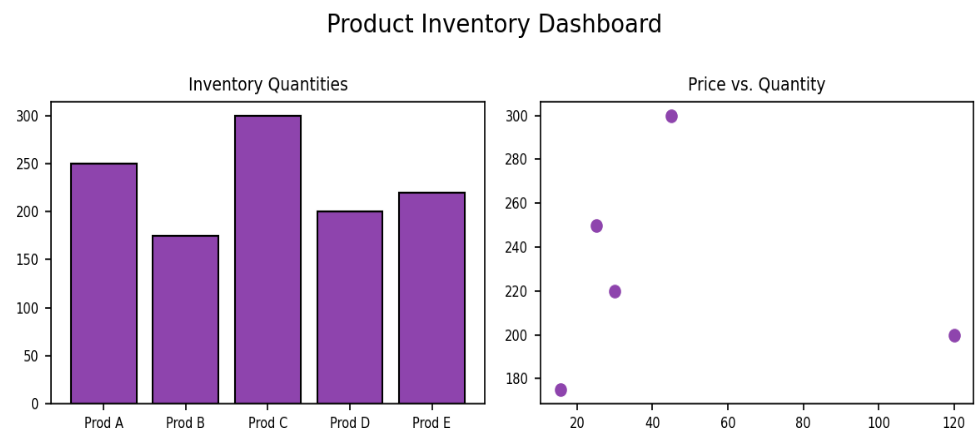
Interpretation: The scatter plot highlights an inverse relationship where lower-priced commodities retain high stock storage volumes (e.g., Product C), whereas high-value luxury stock lines like Product D ($120) maintain tighter, lean inventory footprints.

4. Develop a Tableau dashboard with the bar chart and stacked bar chart to allow users to interact with the data

Tableau Steps:

```
1. Load 'set4_product_inventory.csv' into Tableau data manager.
2. Sheet 1: Columns = Product Name, Rows = Quantity Available to create the
   baseline inventory bar chart.
3. Sheet 2: Columns = Category, Rows = Quantity Available, Color = Product
   Name to split stock across catalog categories.
4. Drop both sheets inside a new grid dashboard container.
```

Dashboard Mock-up:



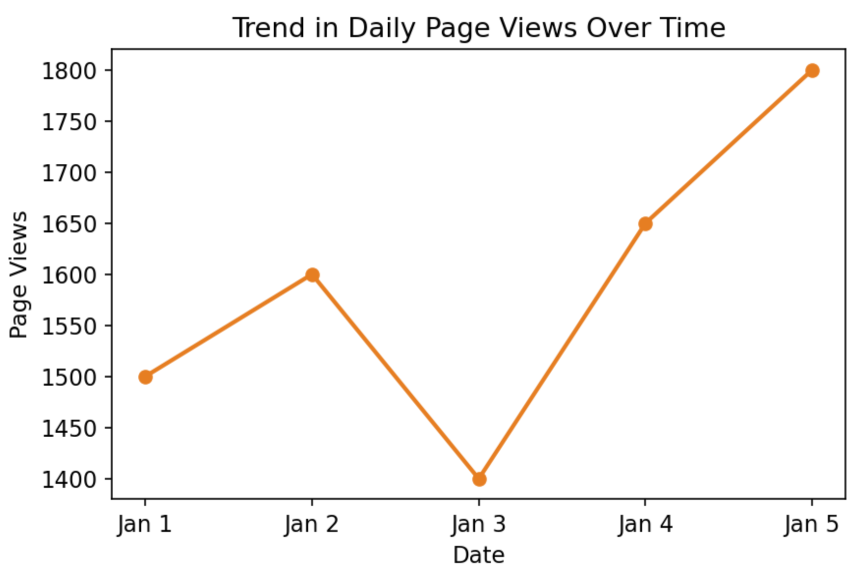
SET 5: WEBSITE ANALYTICS

1. Create a line chart to visualize the trend in daily page views over time

R Code:

```
web_df <- data.frame(
  Date = c('Jan 1','Jan 2','Jan 3','Jan 4','Jan 5'),
  Page_Views = c(1500,1600,1400,1650,1800),
  CTR = c(2.3,2.7,2.0,2.4,2.6))
ggplot(web_df, aes(x=Date, y=Page_Views, group=1)) +
  geom_line(color='#e67e22', size=1) + geom_point() +
  theme_minimal() +
  labs(title='Trend in Daily Page Views Over Time', x='Date', y='Page Views')
```

Output / Chart:

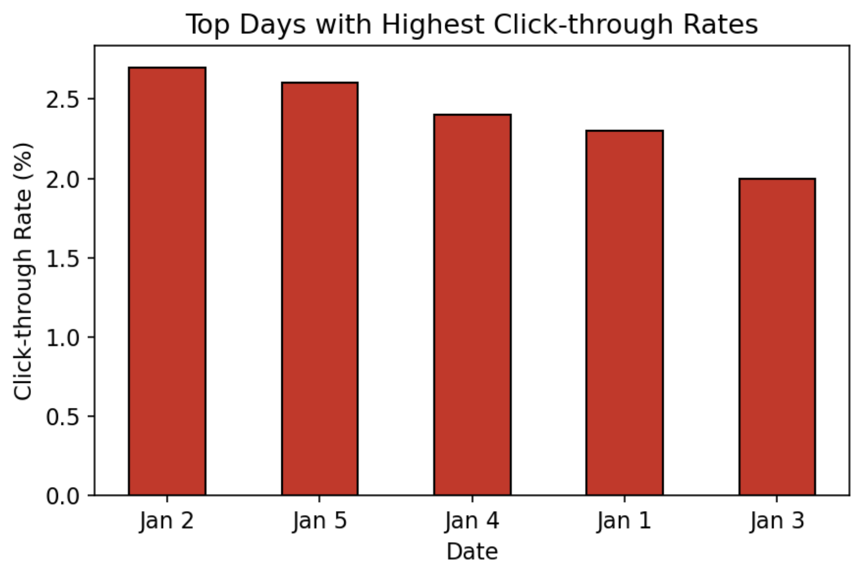


2. Generate a bar chart showing the top N days with the highest click-through rates

R Code:

```
ggplot(web_df, aes(x=reorder(Date, -CTR), y=CTR)) +
  geom_bar(stat='identity', fill='#c0392b', color='black', width=0.5) +
  theme_minimal() +
  labs(title='Top Days with Highest Click-through Rates', x='Date', y='Click-through Rate (%)')
```

Output / Chart:

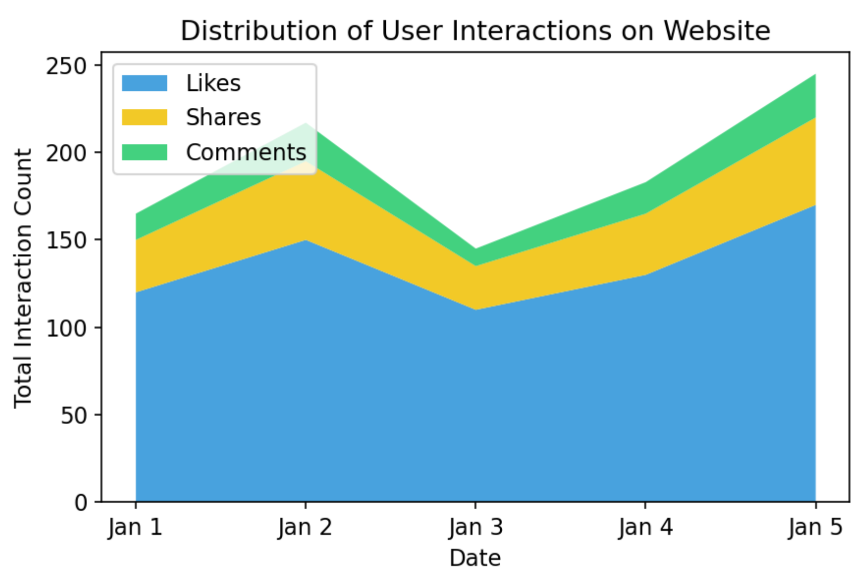


3. Develop a stacked area chart to display the distribution of user interactions on a website

R Code:

```
library(tidyr)
web_df$Likes <- c(120, 150, 110, 130, 170)
web_df$Shares <- c(30, 45, 25, 35, 50)
web_df$Comments <- c(15, 22, 10, 18, 25)
long_web <- gather(web_df, Interaction_Type, Count, Likes:Comments, factor_key=TRUE)
ggplot(long_web, aes(x=Date, y=Count, fill=Interaction_Type, group=Interaction_Type)) +
  geom_area(alpha=0.9) +
  scale_fill_manual(values=c('#3498db','#f1c40f','#2ecc71')) +
  theme_minimal() +
  labs(title='Distribution of User Interactions on Website', x='Date', y='Total Interaction Count')
```

Output / Chart:

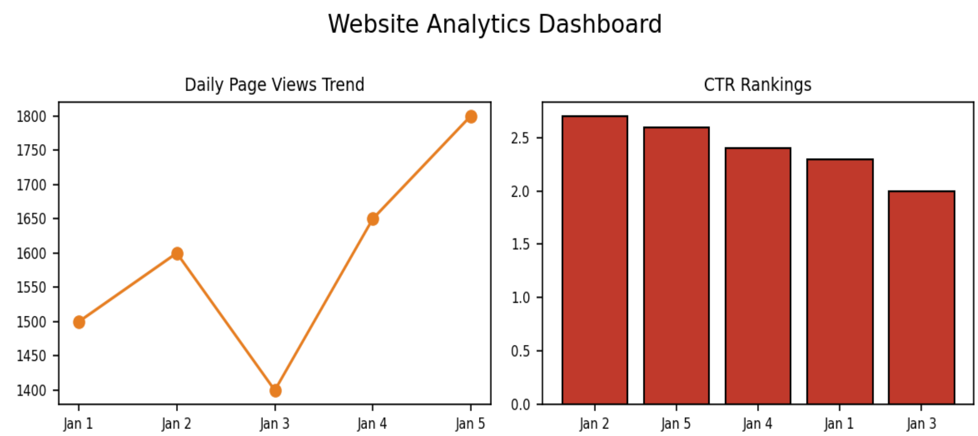


4. Build a Tableau dashboard with the line chart and bar chart for interactive exploration of website traffic data

Tableau Steps:

1. `Connect to data source 'set5_website_traffic.csv'.`
2. `Sheet 1: Date -> Columns, Page Views -> Rows (Line chart execution).`
3. `Sheet 2: Date -> Columns, Click-through Rate -> Rows (Sort descending for Top N days).`
4. `Place both sheets cleanly inside the interactive portal dashboard layout container panel.`

Dashboard Mock-up:



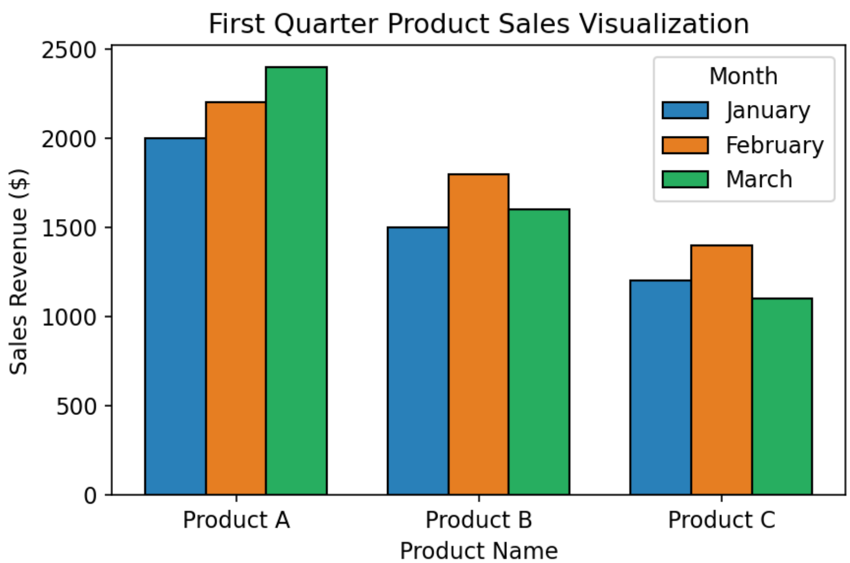
SET 6: PRODUCT SALES ANALYSIS

1. Create a grouped bar chart to visualize the sales of each product for the first quarter

R Code:

```
library(reshape2)
sales_q1 <- data.frame(
  Product_ID = 1:3,
  Product_Name = c('Product A','Product B','Product C'),
  January = c(2000,1500,1200), February = c(2200,1800,1400), March = c(2400,1600,1100))
melted_sales <- melt(sales_q1, id.vars=c('Product_ID','Product_Name'),
                      variable.name='Month', value.name='Sales')
ggplot(melted_sales, aes(x=Product_Name, y=Sales, fill=Month)) +
  geom_bar(stat='identity', position='dodge', color='black') +
  scale_fill_manual(values=c('#2980b9','#e67e22','#27ae60')) +
  theme_minimal() +
  labs(title='First Quarter Product Sales Visualization', x='Product Name', y='Sales Revenue ($)')
```

Output / Chart:

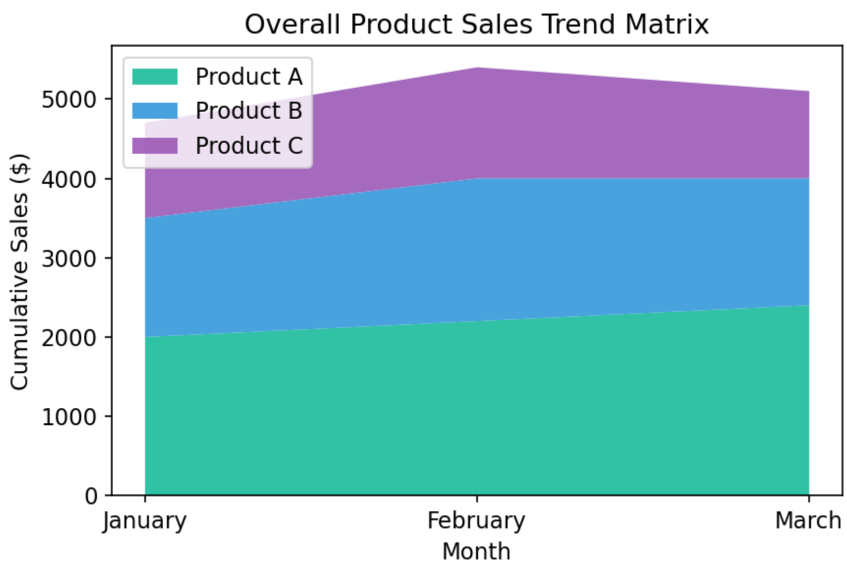


2. Generate a stacked area chart to represent the overall sales trend for all products over the three months

R Code:

```
ggplot(melted_sales, aes(x=Month, y=Sales, fill=Product_Name, group=Product_Name)) +
  geom_area(alpha=0.9) +
  scale_fill_manual(values=c('#1abc9c','#3498db','#9b59b6')) +
  theme_minimal() +
  labs(title='Overall Product Sales Trend Matrix', x='Month', y='Cumulative Sales ($)')
```

Output / Chart:



3. Build a table to show the monthly sales data for each product

R Code:

```
print(sales_q1)
```

Output Screenshot Preview:

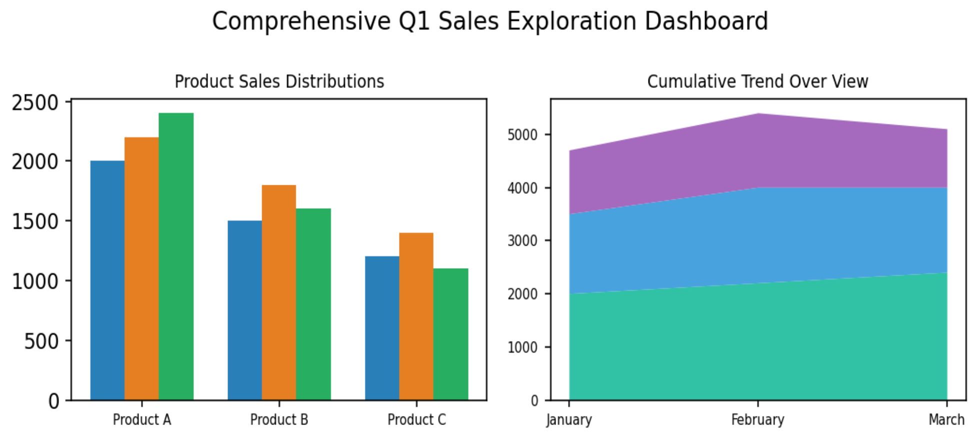
Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	$2,000	$2,200	$2,400
2	Product B	$1,500	$1,800	$1,600
3	Product C	$1,200	$1,400	$1,100

4. Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data

Tableau Steps:

```
1. Load 'set6_monthly_product_sales.csv' and pivot columns (January,
   February, March) into a single month/sales row format structure.
2. Sheet 1: Product Name & Month -> Columns, Sales -> Rows (Grouped view bar
   config).
3. Sheet 2: Month -> Columns, Sales -> Rows, Color = Product Name (Area chart
   transform).
4. Sheet 3: Bring fields as text matrix grid spreadsheet views to form the
   table summary block.
5. Combine all three views inside a unified interactive visual analytics
   workspace report dash.
```

Dashboard Mock-up:


Comprehensive Q1 Sales Exploration Dashboard
Product Sales Distributions
2500
2000
1500
1000
500
0
Product A
Product B
Product C
Cumulative Trend Over View
5000
4000
3000
2000
1000
0
January
February
March